

Amylase Rate, Graph and Topic 1 Consolidation · TA brief

TA Brief

Say once: "I haven't seen what you've been doing... I start from what you show me".

P17 · SLT planning and evaluation

Lesson code: SCI-LAUNCH-A1-W08-L3. SoW reference: LAUNCH Science, Autumn 1 return block; L3.

Exact SoW outcome: I can calculate amylase rate from endpoint time, plot and interpret the pH pattern, evaluate the method, and apply Topic 1 knowledge to exam-style evidence.

Source placement: enzyme activity enrichment consolidates Topic 1; the workbook has no separate Autumn 1 Week 8 row. This is the supplied return-period mapping, not a newly invented workbook objective.

This lesson: I can calculate and graph relative amylase rates, judge what the evidence supports, and use enzyme and transport ideas in an exam-style explanation.

Evidence (scheme of work): Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec). Evidence (this lesson): Show relative-rate working using $1 \div \text{time}$ with the unit s^{-1} ; Read or plot the pH–rate pattern and limit the conclusion to the evidence.; Link a method limitation to an improvement, then distinguish digestion from diffusion.

Evidence location: the pupil's explanation under the book strip, with route, actual support, review edit // and exit response. Keep raw observations distinct from illustrative data.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A1-W08-L3. Record the observed science and prompt level; clicks and access aids do not earn a marking code.

Stage	Minutes	Task	Staff move
Arrival task	3	Choose a route. Think first, then check the answers together.	At arrival, say the spoken line on the marking card once. Use genuine last lesson evidence for the spoken inheritance. This arrival is the only calibration; no separate pre-check or recovery task. IF NEW — 0.5 min read clues; 1.5 min respond; 1 min sample and reveal. IF REVISION — 0.5 min orient; 1.5 min respond; 1 min sample and reveal.

Stage	Minutes	Task	Staff move
Starter · reverse the time	3	Matched amounts, same iodine endpoint: reaction A reaches the endpoint in 90 s; B takes 180 s. Which is faster?	At about minute eight, choose one same-lesson adjustment from the arrival: hold calculator modelling longer, use supported graph placement, switch the current task route or add the repeat-evidence Aspire card. IF NEW — 0.5 min compare times; 1 min predict rates; 1.5 min explain direction. IF REVISION — 0.5 min predict; 1.5 min justify using reciprocals; 1 min check conditions.
I do · time becomes relative rate	4	1 · Identify the measured endpoint time: 120 s.	Use $1/t$ only. Show $1/120 = 0.00833 \text{ s}^{-1}$; discuss sufficient rounding without pretending interval-based observations are exact instantaneous rates. IF NEW — 2.5 min model formula, substitution and unit; 1.5 min connect the result to its graph point. IF REVISION — 0.5 min orient; 2.5 min pupils predict and check calculations; 1 min compare direction.
We do · build an evidence chain	4	Drag a card, or select it then choose its destination.	Named involvement opportunity. Card-then-destination works by keyboard and touch. Ask whether each statement comes from data, biological knowledge or a judgement of the method. IF NEW — 1 min read categories; 2 min classify and justify; 1 min respond to a pupil's reasoning. IF REVISION — 0.5 min orient; 2.5 min classify with evidence; 1 min compare the claim boundary.

Stage	Minutes	Task	Staff move
I do · read the graph honestly	3	pH on x; relative rate / s^{-1} on y. Equal numerical steps have equal spacing.	Label pH 7 as highest in the first series, among values tested. Do not silently average the repeat, impose a universal pH 7 optimum or treat a fall in activity as proof of irreversible denaturation. IF NEW — 1.5 min reveal the five-point pattern; 0.5 min add the separate repeat; 1 min distinguish data from explanation. IF REVISION — 0.5 min inspect; 1.5 min qualify the peak with the repeat; 1 min compare scope and uncertainty.
We do · repair and transfer	4	Draft explanation: "Enzymes are killed at pH 4. pH 7 is the exact optimum for every amylase, and diffusion breaks starch into sugar."	Model one concise repair, then name the two processes. This supplies the Topic 1 transfer before independent work, so a missed earlier topic is not a barrier. IF NEW — 1 min inspect the claim and process definitions; 2 min repair and label; 1 min check. IF REVISION — 0.5 min inspect; 2.5 min repair with a reason; 1 min check the transfer.

Stage	Minutes	Task	Staff move
You do · make the evidence answer	13	Use the supplied evidence. Write, draw, point, explain or direct an adult.	Use pre-scaled axes with three points already shown and two points to add; do not spend the block redrawing axes or copying the full table. Five short responses retain calculation, graph, explanation, evaluation and transfer within fifteen minutes. Supported pupils may direct plotting and dictate the same reasoning. IF NEW — 2 min orient the calculations and graph; 11 min complete the five responses; 2 min use one precise feedback point. IF REVISION — 1 min orient; 12 min complete the five responses; 2 min use one precise feedback point.
Review and improve	3	Keep one correct science claim. Improve one word, unit, piece of evidence or reason. Mark a real change //; read it back if asked.	Give one specific VF and one NS+; wait for a real // edit and hear the pupil explain the change.
Exit ticket	3	Use your book OR one ticket. Show your explanation with the supplied evidence. Say it, write it, draw it, sign it, choose it or show it; an adult can scribe it.	Use one book response OR one ticket. Read once after last lesson to shape the plan for Autumn 2, since half term follows. No retrospective closure or additional pupil recording. IF NEW — 1 min read evidence; 2 min respond; 1 min pupil finishes the same response. IF REVISION — 0.5 min orient; 2.5 min respond; 1 min pupil finishes the same response.

Planned adaptations

Need	Planned adaptation	EEF five-a-day strategy
------	--------------------	-------------------------

Cognition and learning	Arrival: reveal one clue at a time; model a reasoning step, then fade the cue for a fresh example.	scaffolding
Communication and interaction	We do: allow thinking time, paired rehearsal, pointing, sign or AAC before giving a reason.	flexible grouping
SEMH	Independent: agree a quiet pause and one return step; correct privately; accept a pass on a personal example.	cognitive and metacognitive strategies
Sensory and physical	Models: offer reduced motion, paused diagrams, enlarged prepared pieces, seated work or adult-directed placement.	technology
EAL (language access, not SEN)	Arrival: pre-teach topic words beside the diagram; permit any-language planning and a sentence stem for the same science.	explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below. EEF strategies are linked to actual tasks above.

If behind

Must teach core: I can calculate and graph relative amylase rates, judge what the evidence supports, and use enzyme and transport ideas in an exam-style explanation. Shorten or merge: take one shared explanation; preserve independent evidence, review and exit. Next lesson: revisit any insecure criterion before extending the topic. Dependency: Show relative-rate working using $1 \div \text{time}$ with the unit s^{-1} .

Staff evaluation

Date taught	Record actual date.
Delivered as planned	Yes / Partly / No; name change.
Criteria met	Name the criterion and evidence.
Adaptations made and why	Name actual support and effect.
Change from the SoW and coverage impact	State change, reason and next teaching implication.
Evidence banked	Record evidence location and lesson code on the school's evidence app.

P18 · Individual target opportunities

Cognition and learning: Arrival: reveal one clue at a time; model a reasoning step, then fade the cue for a fresh example.

Communication and interaction: We do: allow thinking time, paired rehearsal, pointing, sign or AAC before giving a reason.

SEMH: Independent: agree a quiet pause and one return step; correct privately; accept a pass on a personal example.

Sensory and physical: Models: offer reduced motion, paused diagrams, enlarged prepared pieces, seated work or adult-directed placement.

EAL (language access, not SEN): Arrival: pre-teach topic words beside the diagram; permit any-language planning and a sentence stem for the same science.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan.

Follow the pupil's IBSP where it differs from these routines.

Adult deployment and responsive feedback

At Independent, one adult supports access to Supported while the other holds the group; swap across the week. Wait, gesture, cue, model one step, then try a fresh step independently. Give VF, then one precise NS+; wait for a real pupil response R and // edit. Use ? for read-back. WS and I describe actual support; E requires target-matched evidence.

During We do, the lesson teacher is the named audience. Offer an optional request for reading help, fewer pieces, clearer labels, adult-directed handling or thinking time. Try the change and check its effect. Passing and private requests are valid.

First lesson back · once-only read

Before the lesson, scan the books for five minutes. Notice previous feedback, pupil responses and where the learning stopped.

Ask the cover teacher and TAs what was actually taught, who was present and what changed for individual pupils.

Check the roll against the books. New arrivals have no prior evidence; start from what every pupil shows now.

Existing NS+ is history. Do not chase, back-mark or retrospectively close it. Begin fresh feedback from this lesson.

After the check, one read, once, shapes Autumn 2. Read each success criterion with its access support, choose the next route, then teach; do not reopen earlier feedback.

Written RAG rule and next routes

Show relative-rate working using $1 \div \text{time}$ with the unit s^{-1} . Green: accurate explanation with evidence independently → Aspire application. Amber: part-correct or secure after one content cue → standard with one worked example. Red: no secure evidence after an access adjustment → supported re-model and a fresh check. Record access support separately; do not average criteria.

Read or plot the pH–rate pattern and limit the conclusion to the evidence. Green: accurate explanation with evidence independently → Aspire application. Amber: part-correct or secure after one content cue → standard with one worked example. Red: no secure evidence after an access adjustment → supported re-model and a fresh check. Record access support separately; do not average criteria.

Link a method limitation to an improvement, then distinguish digestion from diffusion. Green: accurate explanation with evidence independently → Aspire application. Amber: part-correct or secure after one content cue → standard with one worked example. Red: no secure evidence after an access adjustment → supported re-model and a fresh check. Record access support separately; do not average criteria.

The criterion profile selects the next routes: begin with the lowest insecure dependency, teach one example, then check fresh evidence and fade the support. A non-response prompts an access check before a science judgement.

Preparation and practical safety

Use the supplied models, cards, data and booklet. This lesson plans only. During the later practical, handle reagents only with the centre-approved method, prepared equipment, local risk assessment and trained supervision. The supplied-observation route needs no reagents and teaches the same interpretation; it does not certify physical practical completion. Do not infer concentrations, heating arrangements or disposal from lesson cards.

Destination link (Launch): rate calculations from timed observations are GCSE Biology exam skills and the everyday work of laboratory technicians, quality-control testers and biomedical science routes at Level 3 and beyond.

Choose a route at each task using what the pupil shows. Switching adds no task, record or label.

Risk-note controls cross-checked against publicly accessible CLEAPSS SSS002 Enzymes and SSS056 Iodine on 26 September 2026. Teacher/technician must authorise the local method, concentrations, quantities, PPE and disposal using current member guidance. The full member-only PP048 method and Hazcards were not accessible for this check.

Use minimum enzyme and iodine concentrations and quantities. Prevent enzyme powder aerosol; do not spray solutions. Follow eye-protection and solvent-specific iodine controls.

Illustrative or supplied data is reasoning practice only and is never logged as practical completion. Log only an observed, authorised physical contribution; the teacher decides whether it meets the relevant course requirement.

Teacher answers and topic guidance

Objective mapping: Retains calculation, graph work, evaluation and Topic 1 transfer; labels 1/time as endpoint-based relative rate and supplies all prerequisite definitions. Fifteen minutes of independent work are preserved.

Real-world link

Teacher answers

Shared activities

Starter · reverse the time: 90 s is faster than 180 s for matched amounts and the same endpoint. $1/90 \approx 0.0111 \text{ s}^{-1}$; $1/180 \approx 0.00556 \text{ s}^{-1}$. Doubling the time halves the relative rate. Worked rate and graph point: Relative rate = $1/\text{time} = 1/120 = 0.00833 \text{ s}^{-1}$. Plot pH 6 at 0.00833 s^{-1} ; pH has no unit. Build an evidence chain: Calculation checks: a reciprocal value; shorter time gives larger rate. Pattern/conclusion: pH 7 has the highest first-series rate among tested values; rate rises to pH 7 then falls at pH 8. Evaluation: colour judgement, shorter sampling intervals, and transparent treatment of repeats/anomalies. Graph and claim repair: First-series rates are 0.00417, 0.00556, 0.00833, 0.0111 and 0.00667 s^{-1} for pH 4–8. The separate pH 7 repeat is 0.00833 s^{-1} , plotted with a distinct symbol at $x = 7$. These values do not prove an exact universal optimum or permanent denaturation. Topic 1 process labels: Starch chemically breaking into smaller sugars is enzyme digestion. Net movement of particles from higher to lower concentration is diffusion. Movement does not itself chemically break starch into sugars.

Supported arrival

1. 90 s is faster.
2. Orange-brown.
3. $1 \div 100 = 0.0100 \text{ s}^{-1}$ (0.01 s^{-1} is also correct).
4. EVALUATE.

Supported independent task

1. pH 6: $1/120 = 0.00833 \text{ s}^{-1}$. pH 7: $1/90 = 0.0111 \text{ s}^{-1}$.
2. Plot (6, 0.00833) and (7, 0.0111), pH on x-axis, relative rate/ s^{-1} on y-axis. Accept equivalent pupil-directed placement.
3. pH 7 is fastest in the first series, because 0.0111 s^{-1} is the highest measured relative rate (or 90 s the shortest time).
4. The exact endpoint between samples is unknown. A shorter fixed interval could narrow the timing window if authorised and feasible; it does not remove every uncertainty.
5. Starch breaking into smaller sugars: digestion. Particles moving down a concentration gradient: diffusion.

Supported exit

1. $1/120 = 0.00833 \text{ s}^{-1}$; pH 7 is faster in the first series (90 s).
2. The first statement is supported. Only pH 4–8 in these conditions were tested; the repeat also shows variation.

Independent arrival

1. The same measured change takes less time; $1/90$ is larger than $1/180$.
2. 90 s, while acknowledging the endpoint was reached after 60 s and by 90 s.
3. Relative rate = $1/120 = 0.00833 \text{ s}^{-1}$.
4. Describe states how values change; explain uses biology, such as pH affecting the active site, to account for the pattern.

Independent independent task

1. Relative rate = $1/\text{time}$. pH 6: $1/120 = 0.00833 \text{ s}^{-1}$. pH 7: $1/90 = 0.0111 \text{ s}^{-1}$. The shorter time produces the larger reciprocal rate.
2. x-axis pH (no unit); y-axis relative rate/ s^{-1} . Points 4:0.00417, 5:0.00556, 6:0.00833, 7:0.0111, 8:0.00667. Rate rises from pH 4 to 7, then falls at pH 8.

3. pH 7 has the highest first-series rate, 0.0111 s^{-1} versus e.g. pH 4 at 0.00417 s^{-1} . pH can alter the active-site environment/shape and affect substrate binding and catalysis. The graph does not prove permanent denaturation or an exact universal optimum.

4. Strength: matched starting amounts/temperature and a common endpoint support comparison. Limitation: 30 s intervals or subjective colour judgement. Improvement: shorter fixed intervals if authorised, or consistent agreed colour judgement; link it to the limitation. The pH 7 repeat gives 120 s, so variation weakens confidence from one series.

5. Digestion is chemical breakdown catalysed by enzymes; amylase breaks starch into smaller sugars. Diffusion is net particle movement from higher to lower concentration. Moving particles does not itself change starch molecules chemically.

Independent exit

1. pH 6: 0.00833 s^{-1} ; pH 7: 0.0111 s^{-1} . pH 7 is faster in the first series under these matched conditions.

2. Variation: repeat matched trials across all pH values to judge the pattern. Or 30 s timing gap: shorter fixed intervals narrow the endpoint window if authorised.

Aspire arrival

1. Its relative rate halves.

2. It summarises time to an observed endpoint, with sampling and colour limits. It does not continuously measure the amount changed per instant.

3. $1/120 = 0.00833 \text{ s}^{-1}$; $1000/120 = 8.33$ on that stated scaled convention. They must not be mixed on one unlabelled axis.

4. For example: one trial per pH limits repeatability evidence, so repeat matched trials before treating the peak as secure.

Aspire independent task

1. pH 6 = $1/120 = 0.00833 \text{ s}^{-1}$. First pH 7 = $1/90 = 0.0111 \text{ s}^{-1}$. Separate repeat pH 7 = $1/120 = 0.00833 \text{ s}^{-1}$; it equals the first pH 6 rate. Keep both pH 7 observations visible.

2. First-series graph uses the five supplied/calculated coordinates. Repeat belongs at (7, 0.00833), as the same IV condition measured again, with its own symbol/key. Do not silently replace or average raw points.

3. The claim is too broad. pH 7 is highest only in the first series among pH 4–8 tested. A 120 s repeat shows variation and matches the first pH 6 time. Untested values and other amylase sources/conditions remain unknown. pH effects on active sites are a biological explanation, not direct molecular observations from this graph.

4. Matched starting amounts/temperature and common endpoint strengthen validity. A 30 s interval limits timing precision; a shorter fixed interval could narrow that gap if feasible/authorised. It does not remove subjective colour judgement, variable starting conditions or all repeat variation.

5. Digestion chemically breaks large molecules down; diffusion moves particles down a concentration gradient. They are different processes. The graph measures an endpoint in starch breakdown, not membrane transport, concentration gradients across a gut wall or transport proteins.

Aspire exit

1. Repeat rate = $1/120 = 0.00833 \text{ s}^{-1}$, matching first-series pH 6. It shows variation and weakens an exact-optimum claim.

2. A shorter authorised interval narrows the timing window, or repeats clarify variation. Lower activity alone does not establish irreversible structural change; the graph does not measure active-site structure or recovery.

OpenStax: enzymes

Next teaching after last lesson: Autumn 2, after half term.

LAUNCH SCIENCE · last lesson · Week 8 · Autumn 2

Staff copy — not for pupil books

Lead with VF in the practical and exit tasks. Written feedback is for extended writing only, strategic and not every piece (Curriculum Policy 2026/27 §14.3).

Core practical evidence. The Edexcel core practical is evidenced by the pupil doing the practical. The illustrative route evidences the interpretation skills only, and is never recorded as core practical completion.

Code

Meaning

Verbal feedback given here

Worked with support

NS + ...

Next step one specific thing

Evidence captured on the school's digital evidence platform

Responded loop closed (Audience has happened)

Self-edit point

Read this back to me

Curriculum Policy 2026/27 §14.3 summary: VF WS I NS E R (six codes; NS "next step", E "evidence on digital software app", R "responded to feedback"). The Feedback Policy, the code source, adds // and ? and expands NS to one specific thing and R to loop closed.

Voice + Audience = R: the pupil responds AND the adult genuinely engages with that response; add R only when both have happened. For a margin edit, the policy says: "in the same colour, on the same page".

Supported: Repair one rate calculation using $1 \div \text{time}$, then check that the shorter time gives the larger rate.

Independent: Add a number and "among the values tested" to one graph conclusion; keep the explanation separate from the observation.

Aspire: Use the separate repeat to limit your optimum claim, then link one method improvement to the uncertainty it actually reduces.

Secure: Use the same evidence-versus-claim discipline in the next taught science week, Autumn 2.

Mixed: Re-work one time-to-rate example and connect it to a single graph point.

Misconception: Contrast a larger time with its smaller reciprocal and separate a tested peak from a universal optimum.

Access barrier: Use pre-scaled axes, a calculator key sequence and pupil-directed point placement.

Method-or-data problem: Keep original and repeat results distinct; re-establish matched amounts, endpoint rule and time units before interpreting.

We do · build an evidence chain. Invite a pupil to classify a rate calculation, graph claim or evaluation statement and justify its role; give feedback on the reason. VF records verbal feedback actually given.

In our amylase investigation lesson, your evidence showed ____, so today we're starting with ____.

LAUNCH: "Adult engages with the edit against the mark scheme / model answer before R." The pupil owns the response; no adult counter-initial or extra receipt. Codes stay in books. No second next step or retrospective closure.

Answer: 90 s is faster.

Answer: Orange-brown.

Answer: $1 \div 100 = 0.0100 \text{ s}^{-1}$ (0.01 s^{-1} is also correct).

Answer: EVALUATE.

Answer: The same measured change takes less time; $1/90$ is larger than $1/180$.

Answer: 90 s, while acknowledging the endpoint was reached after 60 s and by 90 s.

Answer: Relative rate = $1/120 = 0.00833 \text{ s}^{-1}$.

Answer: Describe states how values change; explain uses biology, such as pH affecting the active site, to account for the pattern.

Answer: Its relative rate halves.

Answer: It summarises time to an observed endpoint, with sampling and colour limits. It does not continuously measure the amount changed per instant.

Answer: $1/120 = 0.00833 \text{ s}^{-1}$; $1000/120 = 8.33$ on that stated scaled convention. They must not be mixed on one unlabelled axis.

Answer: For example: one trial per pH limits repeatability evidence, so repeat matched trials before treating the peak as secure.

Science staff guidance

Beside the You do, mark I (independent), S (supported) or C (choice): the level of independence, not the effort.

Give time to start → point to the relevant place → rephrase → offer a small prompt or word bank → reduce to a limited choice. Never straight to telling the answer.

No cold calling; nobody is asked to read difficult text aloud.

Tell the teacher the barrier: reading, vocabulary, processing, regulation, writing or a science misconception. Avoid prompting through every question or dictating the final answer.

E Evidence on the school's evidence app

Lean capture on the school's evidence app (E): 30-second clip, photo, or one-line context note.

Point, sign, repeat after the adult, demonstrate, a clip on the school's evidence app, choice board.

If behind

A pupil who missed last lesson: give the Arrival reminders and today's must-know words, not the missed lesson's work.

Arrival and Starter

Arrival: answer these four short questions in the book.

1. Explain why 90 s gives a larger relative rate than 180 s under matched conditions.
2. At 60 s starch is detected; at 90 s it is not. What endpoint value is used in the rate calculation?
3. Calculate relative rate for an endpoint of 120 s; show the equation, substitution and unit.
4. Distinguish DESCRIBE from EXPLAIN when answering about a graph.

Starter: Matched amounts, same iodine endpoint: reaction A reaches the endpoint in 90 s; B takes 180 s. Which is faster? One short answer in the book.

You do routes

Independent is preferred: work in the book without answer-giving support. Supported uses a small prompt beside the same task. Choice uses limited options, matching or selecting and then a short scientific sentence, label or reason in the book. Aspire is the challenge for every route.

Mark scheme · one point per mark

Independent

1. Calculate relative rate for pH 6 and pH 7, showing equation, substitution and units. Check that shorter time gives a larger value. [4 marks]
 1. Use relative rate = $1/\text{time}$ (1 mark)
 2. $1/120 = 0.00833 \text{ s}^{-1}$ (1 mark)
 3. $1/90 = 0.0111 \text{ s}^{-1}$ (1 mark)
 4. The shorter time gives the larger relative rate (1 mark)
2. Complete the five-point graph using the two calculated rates and three supplied points. Label both axes and state the rise-and-fall pattern. [4 marks]
 1. Plot pH 6 at 0.00833 s^{-1} (1 mark)
 2. Plot pH 7 at 0.0111 s^{-1} (1 mark)
 3. Label pH and relative rate / s^{-1} (1 mark)
 4. The rate rises to pH 7 then falls at pH 8 (1 mark)
3. Identify the highest tested activity in the first series using one numerical comparison. Explain how pH can affect the enzyme's active site. [3 marks]
 1. pH 7 has the highest first-series rate, 0.0111 s^{-1} (1 mark)
 2. Support with another value, such as pH 6 at 0.00833 s^{-1} (1 mark)
 3. pH can change active-site shape or binding and so enzyme activity (1 mark)
4. Give one strength, one limitation and one linked improvement for the endpoint method. Use the separate pH 7 repeat of 120 s to qualify your confidence. [4 marks]
 1. A consistent observable endpoint or matched conditions is a strength (1 mark)
 2. Sampling gaps or subjective colour judgement is a limitation (1 mark)
 3. A shorter authorised interval or agreed colour standard targets that limitation (1 mark)
 4. The pH 7 repeat of 120 s differs from 90 s and limits repeatability confidence (1 mark)
5. Distinguish enzyme digestion from diffusion in the supplied two-step diagram. Explain why movement from high to low concentration does not itself break starch into sugars. [3 marks]
 1. Digestion is chemical breakdown of starch into smaller molecules (1 mark)
 2. Diffusion is net movement down a concentration gradient (1 mark)
 3. Movement itself does not break the starch molecule (1 mark)

Aspire

1. Calculate pH 6 and pH 7 relative rates. Compare the first pH 7 rate with its separate 120 s repeat, retaining both results. [4 marks]
 1. $1/120 = 0.00833 \text{ s}^{-1}$ at pH 6 (1 mark)
 2. $1/90 = 0.0111 \text{ s}^{-1}$ at pH 7 (1 mark)
 3. The separate pH 7 repeat is 0.00833 s^{-1} (1 mark)
 4. Retain both pH 7 trials (1 mark)
2. Complete the five-point first-series graph and place the separate repeat at the same pH 7 x-position with a distinct symbol. Explain why it is not pH 9 or a silently replaced point. [4 marks]
 1. Plot the five first-series points with correct axes (1 mark)
 2. Plot the repeat at pH 7 and 0.00833 s^{-1} (1 mark)
 3. Use a distinct symbol/key (1 mark)
 4. The repeat has the same pH and is not a replacement observation (1 mark)
3. Evaluate “pH 7 is the exact optimum for every amylase”. Use the tested range, the repeat and the difference between data and a molecular explanation. [4 marks]
 1. pH 7 is highest only among the values tested (1 mark)
 2. An exact optimum between tested values was not measured (1 mark)
 3. The repeat shows variation (1 mark)
 4. Molecular mechanism is an explanation, not directly observed in the timing data (1 mark)
4. Give a strength–limitation–improvement chain. Explain which uncertainty a shorter authorised sampling interval reduces and which it leaves. [4 marks]
 1. Identify a relevant strength (1 mark)
 2. Name a sampling-gap limitation (1 mark)
 3. Link a shorter authorised interval to a narrower timing window (1 mark)
 4. It does not remove subjective colour judgement or all systematic error (1 mark)
5. Explain why “diffusion digests starch” confuses two processes. Use the supplied definitions, and state why an enzyme-rate graph alone cannot show how sugars cross a gut membrane. [4 marks]
 1. Digestion changes starch molecules into smaller sugars (1 mark)
 2. Diffusion moves particles down a concentration gradient (1 mark)
 3. The graph measured starch disappearance rather than membrane transport (1 mark)
 4. It cannot establish how sugars cross the gut membrane (1 mark)

Accept scientifically equivalent wording. Award each distinct point once. The mark tariff is for the question, not a participation score. Keep the answer scheme staff-only during independent work.